

# Functional Annotation Report: *paaC* (Q88HS7 / PP\_3276) in *Pseudomonas putida* KT2440

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## Gene/Protein Identity Verification

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- **UniProt:** Q88HS7 — "Ring 1,2-phenylacetyl-CoA epoxidase beta subunit"; EC 1.14.13.149
- **Gene:** *paaC*; OrderedLocusName **PP\_3276**
- **Organism:** *Pseudomonas putida* KT2440 (ATCC 47054 / DSM 6125)
- **Domains (InterPro):** Aromatic\_CoA\_ox/epox. (IPR052703); Ferritin-like (IPR012347); Ferritin-like\_SF (IPR009078); PaaA\_PaaC (IPR007814); PaaC (IPR011882)

**Verification outcome — CONFIRMED.** The gene symbol *paaC*, the "phenylacetyl-CoA epoxidase beta subunit" description, the ferritin-like/PaaC domain signatures, and the EC number all point consistently to the **β (structural) subunit of the bacterial phenylacetyl-CoA oxygenase/epoxidase complex** of the phenylacetic acid (PAA) catabolic pathway. This pathway is present in *E. coli* and *P. putida* and is among the best-characterized aerobic aromatic-degradation systems, so authoritative primary literature is abundant and directly applicable. The nomenclature (PaaA/B/C/D/E) is conserved between *E. coli* and *P. putida*; mechanistic and structural work performed in *E. coli* and *Azoarcus/Thermus* orthologs is transferable to the *P. putida* KT2440 protein.

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## 1. Summary (Answer to the Research Question)

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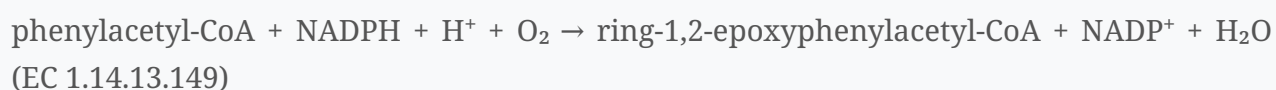
*paaC* encodes the structural **β subunit of the multicomponent phenylacetyl-CoA oxygenase/epoxidase (PaaABC(D)E)**. The complex catalyzes the first committed, ring-activating step of aerobic phenylacetate catabolism: the O<sub>2</sub>- and NADPH-dependent **1,2-epoxidation of the aromatic ring of phenylacetyl-CoA** to form ring-1,2-epoxyphenylacetyl-CoA (EC 1.14.13.149); the same complex can also deoxygenate this reactive epoxide. **PaaC itself is catalytically silent** — the diiron oxygen-activating center resides in the α subunit PaaA —

but PaaC is **structurally indispensable**: it forms the PaaAC catalytic core, scaffolds and stabilizes the PaaA active site, participates in the higher-order PaaAC/PaaACB assembly, and is required for productive electron transfer from the reductase PaaE. The reaction takes place in the **cytoplasm**, acting on a cytoplasmic CoA-thioester substrate.

## 2. Primary Function and Substrate Specificity

### 2.1 The reaction

The PaaABC(D)E complex activates the otherwise inert aromatic ring of **phenylacetyl-CoA** by inserting an oxygen across the C1–C2 bond, producing **ring-1,2-epoxyphenylacetyl-CoA**:



Teufel *et al.* (2010) established that "the aromatic ring of phenylacetyl-CoA becomes activated to a ring 1,2-epoxide by a distinct multicomponent oxygenase," which is then isomerized to a seven-membered oxepin-CoA and hydrolytically cleaved (

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Teufel, Friedrich & Fuchs (2012) biochemically characterized the complex, showing PaaABCE "not only transforms phenylacetyl-CoA into its ring-1,2-epoxide, but also mediates the NADPH-dependent removal of the epoxide oxygen, regenerating phenylacetyl-CoA with formation of water" via "a catalytic di-iron centre" — i.e., it is a **bifunctional oxygenase/deoxygenase** that provides an intrinsic escape from its own toxic epoxide product (

P 22398448

### 2.2 Substrate specificity

The physiological substrate is the **CoA thioester** phenylacetyl-CoA, not free phenylacetate — a defining feature of this "hybrid" aerobic pathway in which all intermediates are CoA-activated (

P 20660314

Crystallographic work with the *E. coli* ortholog captured the enzyme in complex with the true

substrate phenylacetyl-CoA as well as related acyl-CoAs (CoA, 3-hydroxybutyryl-CoA, benzoyl-CoA), showing an active-site route adapted for a CoA-linked substrate

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## 2.3 Role of PaaC specifically

- **PaaA = catalytic  $\alpha$  subunit; PaaC = structural  $\beta$  subunit.** "PaaAC forms a catalytic core with a monooxygenase fold with PaaA being the catalytic  $\alpha$  subunit and PaaC, the structural  $\beta$  subunit" (Grishin *et al.* 2011,

P 21247899

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- **PaaC is required for catalysis.** In vitro reconstitution showed "each of the PaaA, B, C, and E subunits are necessary for catalysis, whereas PaaD is not essential"

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- **Fold/family.** PaaC adopts a **ferritin-like/diiron (bacterial multicomponent monooxygenase-like) fold** but functions as the non-catalytic  $\beta$  partner; the enzyme constitutes "the prototype of a new and distinct group of the large bacterial diiron multicomponent oxygenase family"

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— a "new family of monooxygenases" differing in subunit organization

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This is consistent with the InterPro ferritin-like and PaaA\_PaaC/PaaC domain annotations for Q88HS7.

## 3. Quaternary Structure and Molecular Mechanism

- **Catalytic core.** PaaAC forms heterotetramers "organized very differently from other known multisubunit monooxygenases"

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- **Higher-order assembly.** With PaaB, "the PaaACB complex forms heterohexamers, with a homodimer of PaaB bridging two PaaAC heterodimers" (Grishin *et al.* 2013,

( P 24055609 ).

- **Electron transfer.** The reductase PaaE feeds electrons to PaaA's diiron center; a conserved "lysine bridge" near the Fe site is part of the electron-transfer path, and mutants show 20–50× reduced activity

( P 24055609 ).

PaaC scaffolds this core and thereby enables oxygen activation.

- **Interpretation of PaaC's role.** PaaC lacks a complete iron-liganding set (it is the structural  $\beta$  subunit) but is an integral, obligatory component of the oligomeric catalytic machine: it positions and stabilizes PaaA's diiron active site and contributes to assembly competent for reductase coupling.

## 4. Biological Process and Pathway Context

paaC operates within the **phenylacetic acid "catabolon"**, "the best characterized aerobic hybrid route involved in the bacterial degradation of aromatic compounds" (Fernández *et al.* 2014,

( P 24983528 ).

The pathway:

1. **Activation:** phenylacetate  $\rightarrow$  phenylacetyl-CoA (phenylacetyl-CoA ligase, PaaK/PaaF).
2. **Ring epoxidation (this enzyme):** phenylacetyl-CoA  $\rightarrow$  ring-1,2-epoxyphenylacetyl-CoA (PaaABC(D)E; PaaC =  $\beta$  subunit).
3. **Isomerization:** epoxide  $\rightarrow$  oxepin-CoA.
4. **Ring cleavage:** hydrolytic cleavage by PaaZ (bifunctional oxepin-CoA hydrolase/aldehyde dehydrogenase;

( P 21296885 ).

5.  **$\beta$ -oxidation:** to acetyl-CoA + succinyl-CoA, feeding the TCA cycle

( P 20660314 ).

This route is the **funneling hub** for catabolism of upstream aromatics including **phenylalanine, styrene/ethylbenzene, 2-phenylethylamine, and n-phenylalkanoic acids**, allowing *P. putida* to use them as carbon and energy sources (

P 20660314

The reactive epoxide/aldehyde branch intermediates are also the biosynthetic entry point to tropone/tropolone natural products in some bacteria (

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underscoring the pathway's metabolic importance.

**Regulation in *P. putida* KT2440 (target organism):** paa genes are inducible by PAA and repressed by a PaaX-type regulator responsive to phenylacetyl-CoA (

P 24983528

PAA metabolism is under **carbon catabolite repression** in KT2440, with intracellular 2-keto-3-deoxy-6-phosphogluconate (KDPG) identified as the CCR signal — glucose/gluconate suppress paa expression and phenylacetyl-CoA ligase activity (Kim *et al.* 2009, )

P 19406896

PAA is additionally a **chemoattractant** for *P. putida* F1, sensed by the energy-taxis receptor Aer2 and dependent on active PAA metabolism (

P 23377939

## 5. Subcellular Localization

The enzyme is a **soluble, cytoplasmic** multicomponent oxygenase. Two lines of evidence support cytoplasmic action: (i) the substrate is a **cytoplasmic CoA thioester** (phenylacetyl-CoA), and CoA-activated intermediates are membrane-impermeant metabolites of central carbon metabolism; and (ii) the complex has been purified as soluble subunits/subcomplexes and crystallized (PaaAC, PaaACB), with no membrane-spanning segments reported (

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No signal-peptide or transmembrane features are indicated for the paaC  $\beta$  subunit. Thus paaC carries out its structural role **inside the cytoplasm**, as part of the soluble PAA-degradation machinery.

## 6. Supported and Refuted Hypotheses

**Supported** - *paaC* is the structural  $\beta$  subunit of the phenylacetyl-CoA oxygenase/epoxidase — supported by crystallographic and reconstitution data

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- *The complex epoxidizes the phenylacetyl-CoA ring (EC 1.14.13.149) and can deoxygenate the epoxide* — supported by direct biochemistry

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- *paaC* acts in the cytoplasm within the aerobic PAA catabolon — supported by pathway biochemistry and substrate chemistry

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**Refuted / corrected** - *"Ring hydroxylase" (simple aromatic hydroxylation)* — the older annotation (e.g., "hydroxylates phenylacetyl-CoA,"

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was **superseded**: the true product is a **ring-1,2-epoxide**, not a phenol

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The UniProt "epoxidase" naming reflects this correction. - *PaaC* is *catalytic* — refuted; catalysis (diiron chemistry) resides in **PaaA**, with PaaC serving a structural role (

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## 7. Limitations and Future Directions

- Direct biochemical/structural characterization has been performed largely on *E. coli* (and *Azoarcus/Thermus/Aromatoleum*) orthologs; the specific *P. putida* KT2440 PaaC has been characterized chiefly by genomic/regulatory studies and by orthology, not by its own crystal structure. Given the high conservation of the paa cluster, transfer of function is well justified but not experimentally proven for PP\_3276 in isolation.
- The precise contribution of PaaC residues to diiron-site geometry and to PaaE coupling in *P. putida* would benefit from targeted mutagenesis and a *P. putida*-specific structure (e.g., cryo-EM of PaaABCE).
- Note on nomenclature: some *P. putida* literature (e.g., the Luengo group's *P. putida* U studies) uses alternative gene labels for the oxygenase subunits; care is needed when mapping names across strains, but the KT2440 UniProt annotation (paaC =  $\beta$  subunit) is internally consistent with the *E. coli* PaaABCDE scheme.

## Key References

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**P** 20660314

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**P** 22398448

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P 21247899

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